11th grade
Learning evidence 2.1: Confidence Interval Planning (Practice Solutions)

Evaluated criteria

- C1: Compute the sample mean and sample standard deviation.
- C2: Distinguish between sample statistics and population parameters.
- C3: Explain why interval estimation is preferred over a single point estimate.
- C5: Construct a X% confidence interval using known population standard deviation.
- C4: Interpret the meaning of a X% confidence interval in context.

Problems

P1 Problem description

A distribution center monitors the time (in minutes) required to load outbound pallets. A sample of 75 loading times is grouped into class intervals below. The center has a historical estimate of the population standard deviation, $\sigma = 6{,}0$ minutes. For classroom reference, suppose the true population mean loading time is $\mu = 33$ minutes. The grouped intervals and frequencies are: 18–22 minutes (10), 23–27 minutes (16), 28–32 minutes (19), 33–37 minutes (17), and 38–42 minutes (13). Use the grouped data to estimate the sample mean and sample standard deviation. Construct and interpret a 95% confidence interval for the population mean loading time. Construct and interpret a 98% confidence interval for the population mean loading time.

C1

Using grouped midpoints x with frequencies f.
First, we calculate the sample mean $\overline{x}$

$$n = \sum f \qquad \longrightarrow \qquad n = 10 + 16 + 19 + 17 + 13 = 75$$

$$\bar{x} = \frac{\sum fx}{n} \qquad \longrightarrow \qquad \sum fx = 10(20)+16(25)+19(30)+17(35)+13(40) = 2285 \qquad \longrightarrow \qquad \bar{x} = \frac{2285}{75} \approx 30{,}47$$

Second, we calculate the sample standard deviation s or the sample variance s^2

$$s^2 = \frac{\sum f(x-\bar{x})^2}{n-1} \qquad \longrightarrow \qquad \sum f(x-\bar{x})^2 \approx 10(20-30{,}47)^2+16(25-30{,}47)^2+19(30-30{,}47)^2+17(35-30{,}47)^2+13(40-30{,}47)^2$$

$$\sum f(x-\bar{x})^2 \approx 1095{,}51+478{,}15+4{,}14+349{,}37+1181{,}50 = 3108{,}67 \qquad \longrightarrow \qquad s = \sqrt{\frac{\sum f(x-\bar{x})^2}{n-1}} = \sqrt{\frac{3108{,}67}{75-1}} \approx 6{,}48$$

C2

The sample statistics are $\bar{x} \approx 30{,}47$ and $s \approx 6{,}48$, while the population parameters are μ (the true mean) and $\sigma = 6{,}0$ minutes. The sample values estimate, but do not equal, the population values.

C3

A single point estimate like $\bar{x}$ ignores sampling variability. A confidence interval gives a range of plausible values for μ, which is more informative for planning in a real setting.

C5

With known $\sigma = 6{,}0$ and $n = 75$, the standard error is

$$SE = \frac{\sigma}{\sqrt{n}} \qquad \longrightarrow \qquad SE = \frac{6{,}0}{\sqrt{75}} \approx 0{,}69$$

For 95 % confidence, $z_{0{,}025} = 1{,}96$.

$$E = z_{0{,}025} \cdot SE \qquad \longrightarrow \qquad E = 1{,}96(0{,}69) \approx 1{,}36$$

$$\mu \in \bar{x} \pm E \qquad \longrightarrow \qquad \mu \in 30{,}47 \pm 1{,}36 = [30{,}47 - 1{,}36, 30{,}47 + 1{,}36] = [29{,}11, 31{,}82]$$

For 98 % confidence, $z_{0{,}01} = 2{,}33$.

$$E = z_{0{,}01} \cdot SE \qquad \longrightarrow \qquad E = 2{,}33(0{,}69) \approx 1{,}61$$

$$\mu \in \bar{x} \pm E \qquad \longrightarrow \qquad \mu \in 30{,}47 \pm 1{,}61 = [30{,}47 - 1{,}61, 30{,}47 + 1{,}61] = [28{,}85, 32{,}08]$$

C4

We are 95 % confident the true mean loading time is between about 29.11 and 31.82 minutes, and 98 % confident it is between about 28.85 and 32.08 minutes. The 98 % interval is wider because higher confidence requires a larger margin of error.

P2 Problem description

A school cafeteria tracks the number of sandwiches sold during lunch to plan inventory. A random sample of 20 lunch periods is recorded below. The district provides a historical population standard deviation of $\sigma = 5{,}5$ sandwiches.

- Sample ($n = 20$): 42, 38, 45, 47, 40, 44, 39, 41, 46, 43, 37, 48, 40, 45, 42, 44, 39, 46, 41, 43.

Use the sample to compute the mean and sample standard deviation. Construct and interpret a 95 % confidence interval for the population mean sandwiches sold. Construct and interpret a 98 % confidence interval for the population mean sandwiches sold.

C1

First, we calculate the sample mean $\overline{x}$

$$\bar{x} = \frac{\sum x}{n} \quad \longrightarrow \quad \sum x = 42+38+45+47+40+44+39+41+46+43+37+48+40+45+42+44+39+46+41+43 = 850$$

$$\longrightarrow \quad \bar{x} = \frac{850}{20} = 42{,}5$$

Second, we calculate the sample standard deviation s or the sample variance s^2

$$s^2 = \frac{\sum(x-\bar{x})^2}{n-1} \quad \longrightarrow \quad \sum(x-\bar{x})^2 =$$
$$2(42-42{,}5)^2 + 2(40-42{,}5)^2 + 2(45-42{,}5)^2 + 2(44-42{,}5)^2$$
$$+\,2(39-42{,}5)^2 + 2(46-42{,}5)^2 + 2(41-42{,}5)^2 + 2(43-42{,}5)^2$$
$$+\,(38-42{,}5)^2 + (47-42{,}5)^2 + (37-42{,}5)^2 + (48-42{,}5)^2 = 185$$

$$s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}} \quad \longrightarrow \quad s = \sqrt{\frac{185}{20-1}} = \sqrt{\frac{185}{19}} \approx 3{,}12$$

C2

The sample statistics are $\bar{x} = 42{,}5$ and $s \approx 3{,}12$, while the population mean μ and population standard deviation $\sigma = 5{,}5$ are parameters for all lunch periods. The sample measures are estimates of the population values.

C3

Because only 20 lunch periods are observed, a point estimate may be off. Confidence intervals quantify that uncertainty and show a plausible range for μ instead of a single guess.

C5

With known $\sigma = 5{,}5$ and $n = 20$,

$$SE = \frac{\sigma}{\sqrt{n}} \quad \longrightarrow \quad SE = \frac{5{,}5}{\sqrt{20}} \approx 1{,}23$$

For 95 % confidence, $z_{0{,}025} = 1{,}96$.

$$E = z_{0{,}025} \cdot SE \quad \longrightarrow \quad E = 1{,}96(1{,}23) \approx 2{,}41$$

$$\mu \in \bar{x} \pm E \quad \longrightarrow \quad \mu \in 42{,}5 \pm 2{,}41 = [42{,}5 - 2{,}41, 42{,}5 + 2{,}41] = [40{,}09, 44{,}91]$$

For 98 % confidence, $z_{0{,}01} = 2{,}33$.

$$E = z_{0{,}01} \cdot SE \quad \longrightarrow \quad E = 2{,}33(1{,}23) \approx 2{,}87$$

$$\mu \in \bar{x} \pm E \quad \longrightarrow \quad \mu \in 42{,}5 \pm 2{,}87 = [42{,}5 - 2{,}87, 42{,}5 + 2{,}87] = [39{,}63, 45{,}37]$$

C4

We are 95 % confident the true mean sandwiches sold per lunch period is between about 40.09 and 44.91, and 98 % confident it is between about 39.63 and 45.37. The wider 98 % interval offers more assurance at the cost of precision.